Technology in Society 82 (2025) 102905



Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc





# How to realize digital knowledge innovation through digital technology? A perspective based on knowledge digitization and inter-organizational knowledge sharing

Qiang Cheng *, Chun Peng, Huaxin Wan, Yuanhui Dai, Shuangyang Zhang

*School of Economics and Management, Southwest Petroleum University, Chengdu, China*

ARTICLE INFO

*Keywords:*
Digital technology
Knowledge digitization
Inter-organizational knowledge sharing
Digital knowledge innovation

ABSTRACT

This research investigates the mechanisms through which digital technology (DT) influences digital knowledge innovation (DKI), and the roles of knowledge digitization (KD) and inter-organizational knowledge sharing (IKS). The sample consists of 298 members from digital groups in China who participate in inter-organizational collaboration and collaborative innovation activities. Grounded in social interaction theory, this paper uses partial least squares structural equation modeling (PLS-SEM) to examine the relationship between digital technology, knowledge digitization, inter-organizational knowledge sharing, and digital knowledge innovation. The results show that digital technology has a significant positive effect on digital knowledge innovation. Furthermore, digital technology positively impacts digital knowledge innovation by enhancing both knowledge digitization and inter-organizational knowledge sharing. Meanwhile, knowledge digitization and inter-organizational knowledge sharing play a significant serial mediating role between the two. These findings provide guidance for organizations to utilize digital technology, promote knowledge digitization, accelerate knowledge innovation, and enhance knowledge heterogeneity and organizational competitiveness.

## 1. Introduction

In the era of globalization and digitization, inter-organizational relationships are becoming more complex, and competition among organizations is increasingly fierce. To achieve and sustain a competitive advantage, organizations require a continuous influx of knowledge. One approach is to concentrate on internal knowledge generation and dissemination (Antonelli et al., 2022), that is, producing new knowledge or replacing existing knowledge through knowledge creation. However, knowledge creation does not inherently include the application of knowledge. Conversely, knowledge innovation encompasses both the generation of novel knowledge and its subsequent transformation into tangible outcomes via commercialization or application (Rehman et al., 2021), consistently focusing on new value. Because knowledge innovation is crucial for sustained organizational growth and competitiveness, organizations should develop and strengthen their knowledge innovation capabilities. This will enhance their agility and adaptability, ultimately improving their competitive position.

Knowledge innovation requires a diversity of knowledge sources, which facilitates the digital transformation of traditional processes (Kohli & Melville, 2019). However, knowledge barriers exist due to the complex environment and scarcity of knowledge, making it difficult for organizations to access external knowledge (Ravichandran & Giura, 2019). Traditional knowledge innovation theories are insufficient to address current innovation challenges. Nambisan et al. (2017) propose the paradigm of digital innovation management, emphasizing the use of digital technologies to improve innovation processes and enhance innovation effectiveness.

Digital knowledge innovation is the process of leveraging emerging technologies to reconfigure and optimize existing knowledge, thereby generating new value (Maione et al., 2024). This new value manifests in various ways, including increased efficiency in knowledge production and utilization, reduced innovation costs, development of new products and services, improved customer experiences, and the catalysis of new business models. Effective knowledge management promotes knowledge dissemination, increasing the organization's knowledge density and subsequently fostering innovation initiatives. The openness and multifaceted nature of digital technologies offer opportunities for digital knowledge management. On the one hand, digital technologies make knowledge easily storable, transferable, and processable. This

* Corresponding author.
*E-mail addresses:* chengqiang001@swpu.edu.cn (Q. Cheng), pengchun0218@gmail.com (C. Peng).

https://doi.org/10.1016/j.techsoc.2025.102905
Received 11 October 2023; Received in revised form 9 April 2025; Accepted 10 April 2025
Available online 11 April 2025
0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

transforms traditional knowledge management and promotes knowledge digitization, which involves using advanced technologies (e.g., information systems) to encode knowledge into digital formats for storage, computation, and analysis (Cheng et al., 2023). Digital knowledge transcends organizational boundaries, enabling real-time knowledge exchange. This promotes knowledge sharing, collaboration, and inter-organizational knowledge flow. Inter-organizational knowledge exchange integrates diverse resources and experiences, fostering collaborative innovation. On the other hand, social interaction theory emphasizes the mutual influence and interdependence of individuals. Individual behavioral decisions are shaped not only by the organizational environment but also by the dynamics of interpersonal relationships. In the context of digitization, this interconnection is amplified, particularly during knowledge sharing and collaborative innovation. These processes involve interactions among individuals, organizations, and their environment, creating a complex and dynamic knowledge ecosystem (Durlauf & Ioannides, 2010). Therefore, knowledge innovation is not a unidirectional process, but rather a collaborative interaction among multiple agents within a complex network. Organizations foster close connections through digital technologies. They utilize social media, online communities, and open innovation platforms to facilitate the rapid dissemination and evolution of knowledge.

Digital knowledge innovation has emerged as a field of increasing significance in recent years. The rapid advancement of digital technologies has provided convenient and efficient avenues for knowledge innovation. This has transformed knowledge exchange both within and across organizations (Agostini et al., 2019), enhancing collaboration and fostering cooperative innovation. Existing research has recognized the significant impact of the digital environment on a company's innovative capabilities (Gao et al., 2023). However, much research focuses on a single organizational perspective. Exploring knowledge innovation solely from within an organization is insufficient to secure long-term competitive advantages. Therefore, it is imperative to examine the necessity and benefits of inter-organizational knowledge sharing, which is defined as the exchange of knowledge resources between different organizations to help overcome geographical and temporal limitations, foster mutual learning and cooperation, and enhance both innovative capacity and competitive advantage (Chen et al., 2014). Specifically, knowledge digitization encompasses both internal and external knowledge interactions. It is closely related to knowledge sharing and innovation, significantly influencing a company's innovative performance (Han et al., 2024). Therefore, this study aims to explore how digital technology facilitates digital knowledge innovation through knowledge digitization and inter-organizational knowledge sharing, addressing a gap in the existing literature.

To address these issues, this study constructs a theoretical framework to explore how digital technology influences digital knowledge innovation. Specifically, this paper uses PLS-SEM to develop a serial mediation model. This model examines the interplay between knowledge digitization and inter-organizational knowledge sharing as influenced by digital technology, and their combined roles in facilitating digital knowledge innovation.

This study makes significant theoretical and practical contributions to the fields of knowledge digitization and digital knowledge innovation. First, by exploring the impact of digital technology on digital knowledge innovation, this study enriches knowledge innovation theory within a digital context. This lays a foundation for future theoretical and empirical research. Second, this research validates the mediating effects of knowledge digitization and inter-organizational knowledge sharing on the relationship between digital technology and digital knowledge innovation. This elucidates the underlying mechanisms, provides new empirical insights, and broadens our understanding of knowledge digitization and inter-organizational knowledge sharing as crucial facilitators of knowledge innovation in a digital environment.

## 2. Literature review and theoretical foundation

### 2.1. Literature review

Digital technologies propel information flow via data streams, which in turn fosters knowledge accumulation and creation, establishing a novel paradigm for knowledge innovation within broad ecosystems (Xia & Johar, 2025). This dynamic and open nature significantly improves knowledge dissemination and collaborative innovation across organizational boundaries (Zou et al., 2022). Xie et al. (2022) posit that digital platforms facilitate enhanced user engagement, thereby promoting customized innovation. Simultaneously, digitization promotes technology spillovers among enterprises, fostering a more open and collaborative innovation environment, which enables more efficient knowledge integration and innovation (Du & Wang, 2024). Regarding indirect effects, digital technologies can: (1) accelerate technological innovation both by enhancing knowledge reconstruction capabilities (Ciarli et al., 2021) and through collaborative innovation and knowledge spillover (Liu, 2025); (2) positively influence corporate innovation by facilitating knowledge generation, flow, and absorption (Chen et al., 2024); (3) improve knowledge fluidity and innovation efficiency by optimizing information processing workflows and reducing knowledge storage costs (Chen & Kim, 2023).

While many studies highlight the positive impact of digital technologies on knowledge innovation, some research suggests that overreliance may hinder creativity and reduce innovative capacity. Overreliance on digital technologies, particularly algorithms and templates, can limit companies' independent thinking and problem-solving capabilities when addressing complex issues (Orlando et al., 2020). Furthermore, digitization can disperse innovation resources and factors, potentially decreasing research and development (R&D) efficiency (Chen et al., 2023). It is evident that if companies fail to manage digital technologies effectively, the driving effects of these technologies on innovation may encounter numerous challenges.

Despite existing research exploring the impact of digital technology on knowledge innovation and achieving certain outcomes, there remain several pressing issues that require further investigation. Firstly, the current literature primarily focuses on the application and outcomes of digital technology within innovation practices; however, only a few studies side or qualitatively describe their effects on knowledge innovation. The causal relationship between digital technology and knowledge innovation has yet to be fully substantiated. Secondly, the mechanisms underlying the relationship between digital technologies and knowledge innovation remain poorly understood, particularly the key mediating mechanisms. A significant gap remains in the empirical research and theoretical frameworks explaining the specific mechanisms by which digital technology influences knowledge innovation through its impact on knowledge management processes. This study uses cross-sectional data from Chinese digital groups to address these questions, offering theoretical guidance and practical insights for companies seeking to leverage digital technologies for knowledge innovation.

### 2.2. Social interaction theory

This study develops an analytical framework based on social interaction theory. Central to social interaction theory is the bidirectional characteristics of interaction (Blumer, 1986). This theory posits that knowledge creation and innovation are not solitary endeavors, but result from interactions among individuals, between individuals and groups, and among groups (Nonaka & Takeuchi, 1995). Through the exchange of symbolic information, participants continuously adapt their behavioral strategies and reconstruct their cognitive frameworks within specific contexts, thus fostering iterative knowledge innovation (Mead et al., 2015).

In digital environments, social interactions among enterprises follow two distinct evolutionary paths. First, digital technologies enhance the

frequency and effectiveness of inter-organizational interactions by reshaping the interaction landscape. Technologies like blockchain and cloud computing eliminate geographical limitations (Agostini et al., 2019), and algorithm-based trust mechanisms lower transaction costs (Zhou et al., 2022). This facilitates the digital transfer and reorganization of knowledge. Second, the topological structure of inter-organizational network relationships directly impacts knowledge innovation performance. Digital connectivity within business ecosystems allows companies to move beyond traditional resource dependency, integrating diverse knowledge through dynamic alliances (Di Vaio et al., 2021).

Moreover, knowledge digitization serves as a key technological mediator in social interactions. Digitizing tacit knowledge into structured data assets significantly improves its transferability and reconfigurability (Takko et al., 2023). This significantly expands the scope of inter-organizational knowledge sharing. Inter-organizational knowledge sharing, a crucial behavioral element, creates innovation value through external social interaction networks. Digital technology-enabled interactive networks effectively amplify knowledge spillovers (Cabrilo et al., 2024). This enhanced interaction fosters new ideas and facilitates knowledge integration and reconfiguration, thus driving knowledge innovation.

Therefore, social interaction theory offers a critical theoretical lens for understanding knowledge innovation in digital environments. Digital technologies significantly enhance the frequency, effectiveness, and scope of social interactions by reshaping interaction landscapes, enabling knowledge digitization, and strengthening network relationships. This enhanced social interaction fosters conditions conducive to knowledge acquisition, integration, and application, thus profoundly influencing digital knowledge innovation.

## 3. Hypotheses development

### 3.1. Digital technology and digital knowledge innovation

Social interaction theory posits that knowledge exchange is the foundation of innovation. The acquisition, integration, and application of knowledge depend on interactions among individuals and organizations. Through these interactions, organizations share knowledge, experiences, and insights, fostering the development of new ideas and solutions. Digital technologies significantly propel the democratization of innovation by enabling frequent and broad knowledge exchange across organizational boundaries (Švarc et al., 2024). In the knowledge acquisition phase, digital technologies expand the channels and scope of knowledge gathering. Big data technologies allow enterprises to analyze large datasets, revealing market trends and consumer needs, thus gaining a more comprehensive and in-depth understanding of the market (Rahul et al., 2023). Concurrently, cloud computing provides substantial data storage and computational power, enabling rapid knowledge processing and analysis, thereby improving the efficiency of knowledge acquisition (Gupta et al., 2023).

During knowledge integration, digital technologies facilitate the merging and restructuring of knowledge (Ghasemaghaei et al., 2018). Social media platforms foster interactions among enterprises, external experts, partners, and customers, facilitating the integration of internal and external knowledge (Appio et al., 2016). Online collaboration platforms support knowledge sharing and collaboration among team members, enhancing the conversion of tacit knowledge into explicit knowledge (Jasimuddin et al., 2019). This integration of knowledge across organizational boundaries creates a more comprehensive and diverse knowledge base.

In the knowledge application phase, digital technologies accelerate the conversion of knowledge into innovative results. Enterprises utilize acquired and integrated knowledge to improve organizational structures, processes and production methods (Bouncken et al., 2021), thus converting knowledge into tangible products, services, or process innovations (Otioma, 2023). Based on this premise, we propose the following hypothesis:

**Hypothesis 1.** Digital technology has a positive impact on digital knowledge innovation.

### 3.2. The mediating role of knowledge digitization

Digital technology makes knowledge plastic, editable and interactive, thereby reducing the costs of knowledge management while enhancing the efficiency of knowledge utilization. For example, the Internet of Things (IoT), blockchain and cloud computing can transform low-level entity knowledge into digital information and data. This substantially reduces the resources required to search, store, assimilate, and employ knowledge (Antonelli, 2017), facilitating knowledge reuse. Ardolino et al. (2018) show that enterprises can collect real-time data during production by deploying IoT devices, converting this data into actionable knowledge to optimize production workflows.

Moreover, digital platforms and online communities facilitate knowledge sharing and collaboration among diverse organizations and individuals, removing geographical and organizational barriers (Volz et al., 2025). This efficient knowledge exchange optimizes resource allocation within organizations, allowing different types of knowledge to be more effectively integrated and applied. For example, companies can use open innovation platforms like InnoCentive to present challenges, attracting global experts, developers, and users to discuss, contribute ideas, and provide feedback, fostering the development of new digital solutions. This open, boundaryless social interaction can facilitate knowledge integration and innovation by incorporating diverse external knowledge and accelerating knowledge flow (Dabić et al., 2023).

Crucially, digital technologies facilitate dynamic management and updating of knowledge. Big data and artificial intelligence technologies can process and analyze large datasets in real-time, extracting valuable knowledge and providing feedback to organizations (Ciarli et al., 2021). This allows enterprises to better understand market changes and respond rapidly. Pellizzoni et al. (2020) further emphasize that digital technology-supported knowledge management systems facilitate continuous knowledge updating and iteration, ensuring organizations remain current with the latest insights. This dynamic knowledge updating mechanism not only improves information timeliness and accuracy but also fosters continuous knowledge accumulation and application. For instance, by analyzing customer behavior data with machine learning algorithms, enterprises can gain instant insights into changing customer demands and translate these insights into digital products or services (Carbo-Valverde et al., 2020). Based on this, we propose the hypothesis:

**Hypothesis 2a.** Digital technology has a positive impact on knowledge digitization.

Knowledge digitization transforms fragmented, tacit knowledge into explicit, codifiable forms, enabling more frequent and broader social interactions via digital platforms like enterprise social networks and knowledge management systems. This efficient interaction extends beyond organizational boundaries, enhancing knowledge flow both internally and externally (Cheng et al., 2023), and significantly improving the efficiency of knowledge storage, retrieval, and sharing (Han et al., 2024). This is evident not only in knowledge sharing and exchange but also in the in-depth exploration of large digital knowledge datasets. By lowering knowledge acquisition costs, it provides a broader knowledge base for digital innovation. Ilvonen et al. (2018) assert that digitization is a key driver of knowledge innovation, enhancing knowledge accessibility and usability, thereby strengthening organizational innovation capacity.

Knowledge digitization has expanded the avenues for knowledge integration through the establishment of knowledge repositories,

knowledge graphs, and other forms (Jiménez-Castillo & Sánchez-Pérez, 2013). This not only facilitates knowledge exchange and collaboration among employees from various departments and backgrounds within the organization but also enables easier integration of external knowledge resources. Such cross-disciplinary and inter-organizational knowledge integration and reconfiguration allow for more efficient acquisition and value assessment of external knowledge (Lepore et al., 2022), accelerating knowledge internalization (Martínez-Caro et al., 2020). This efficient knowledge absorption and internalization mechanism shortens the time from acquisition to application, fostering innovative solutions and providing renewed impetus for digital knowledge innovation. Thus, we propose a hypothesis:

**Hypothesis 2b**. Knowledge digitization has a positive impact on digital knowledge innovation.

Bouncken et al. (2021) contend that digital technologies form the cornerstone of knowledge digitization. Digital technologies expand the breadth of knowledge within enterprises by facilitating resource aggregation, opportunity recognition, and knowledge sharing, thus stimulating their innovative capabilities (Gao et al., 2023). Hussain et al. (2024) further emphasize that electronic knowledge, as a vital form of knowledge, significantly enhances organizational innovation performance by integrating information and optimizing knowledge management. Specifically, organizations accelerate the integration of R&D resources and knowledge through digital transformation, a process reliant on knowledge digitization capabilities. For example, through data intelligence analysis and live information sharing, organizations can more effectively leverage both internal and external knowledge to drive innovation.

Moreover, the codability of knowledge digitization enables organizations to transform customer feedback, market research reports, patent documents, and other resources into structured data, providing robust data support for product innovation (Bednar & Spiekermann, 2024). The convergence of knowledge digitization promotes feedback recognition across departments and organizations, optimizing the product iteration process. Finally, by analyzing production process data, organizations can identify potential efficiency bottlenecks and implement process improvements. This knowledge generativity empowers enterprises to sustain innovation and maintain a competitive advantage. Accordingly, we hypothesize that:

**Hypothesis 2c**. Knowledge digitization plays a mediating role between digital technology and digital knowledge innovation.

### 3.3. The mediating role of inter-organizational knowledge sharing

Social interaction theory emphasizes that knowledge sharing among organizations relies not only on information transmission but also on the relationship between interacting parties, the nature of the interaction, and the level of trust (Nahapiet & Ghoshal, 1998). Digital technologies have reshaped social interaction patterns, providing organizations with diverse channels for knowledge exchange (Ahmed et al., 2019), reducing costs and improving timeliness associated with external communication and coordination. For instance, IoT facilitates the creation of shared databases between organizations (Ting et al., 2020). This not only reduces knowledge acquisition and transfer costs but also strengthens inter-organizational connections by providing a shared information space, promoting transparency, and fostering trust among organizations. Social networks have become crucial interactive platforms for acquiring information and ideas remotely (Shwartz-Asher et al., 2020). They broaden the scope of social interactions among organizations, enabling connections beyond geographical and organizational boundaries, thus forming a more extensive network of relationships (Chatterjee et al., 2021).

Historically, geographical location has been a significant barrier to knowledge exchange between organizations and their external environments. It restricts face-to-face interactions, which are considered vital for building relationships and conveying tacit knowledge (Ryan & O'Connor, 2013). Digital technologies effectively mitigate the spatio-temporal barriers between organizations, weakening geographical constraints and reducing collaboration costs across distances (Forman & van Zeebroeck, 2019). By providing virtual interaction spaces, digital technologies offset interaction deficiencies caused by physical distance, making remote collaboration and knowledge sharing increasingly common.

Moreover, digital technologies facilitate more efficient knowledge flow (Boeker et al., 2021). Social interaction theory posits that frequent and effective interactions strengthen inter-organizational relationships, reduce cognitive gaps, and enable widespread knowledge dissemination. Instant messaging and video conferencing enhance interaction frequency and feedback speed, helping establish and maintain trust among organizations, which, in turn, facilitates knowledge sharing. Digital technologies have also altered the scope and speed of information acquisition, increasing collaboration frequency among organizations and strengthening the connectivity and stability of inter-organizational knowledge exchanges (Antonelli, 2017), thus promoting knowledge and resource complementarity. Based on this, we propose the hypothesis:

**Hypothesis 3a**. Digital technology has a positive impact on inter-organizational knowledge sharing.

Knowledge sharing is fundamental to innovation, with a primary objective of fostering new ideas and addressing challenges (Azeem et al., 2021). Knowledge sharing is not merely information transmission; it is also a process of interaction and collision among individuals or organizations with diverse knowledge backgrounds. This engagement can spark new perspectives and facilitate the reinterpretation and recombination of existing knowledge, fostering innovative ideas (Nonaka & Takeuchi, 1995). An increasing number of organizations are actively embracing open innovation, assimilating internal ideas with external insights (Chesbrough, 2020). This highlights that inter-organizational knowledge sharing has become a vital component of contemporary enterprises' innovation strategies. Organizations enhance their innovative capabilities by assimilating external knowledge to fill gaps in their knowledge systems. As organizations interact with the external environment and embrace new insights, learning and collaboration naturally follow. Digital knowledge sharing, through knowledge networks, provides organizations with diverse knowledge, promotes cross-fertilization and complementarity of knowledge between organizations (Ozdemir & Gupta, 2021). This not only broadens knowledge sources but also enhances knowledge flow quality, narrowing the knowledge gap among organizations and creating favorable conditions for knowledge innovation.

Furthermore, digital technology has transformed inter-organizational knowledge sharing and innovation from unidirectional transmission to bidirectional flow. This enables continuous knowledge renewal and iteration. The collaborative environment fostered by knowledge sharing enables multilateral relationships, promoting mutual influence among network participants (Zhou et al., 2025). Such collaboration harnesses new technologies and solutions to jointly develop new products, technologies, or services. This reduces resource allocation redundancy, enhances creative efficiency (Ozdemir & Gupta, 2021), and ultimately drives the advancement of digital knowledge innovation. Based on this, this paper proposes the hypothesis:

**Hypothesis 3b**. Inter-organizational knowledge sharing has a positive impact on digital knowledge innovation.

Digital technology has significantly enhanced the efficiency and scope of inter-organizational knowledge sharing (Boeker et al., 2021). This inter-organizational knowledge transfer lays the groundwork for knowledge integration and innovation. Existing research indicates that knowledge sharing is a crucial prerequisite for knowledge innovation,

and the frequency and diversity of knowledge exchange among organizations directly influence its effectiveness (Tønnessen et al., 2021). In digital environments, organizational boundaries become increasingly blurred, enabling more seamless interaction and integration of specialized knowledge and participants from diverse fields (Li et al., 2025). For example, through social media, enterprises can achieve more efficient cross-departmental communication and knowledge integration, accelerating new product development (Sundaresan, Zhang & Justin, 2020). Moreover, by utilizing digital methods to enhance collaboration with external entities (e.g., research institutions and information intermediaries), companies can significantly bolster their knowledge innovation capacity (Zou et al., 2022). Therefore, the hypothesis is proposed in this study:

**Hypothesis 3c**. Inter-organizational knowledge sharing plays a mediating role between digital technology and digital knowledge innovation.

### *3.4. The serial mediating role of knowledge digitization and inter-organizational knowledge sharing*

Digitalization has strengthened proximity between organizations, enabling them to more easily search for, acquire, and integrate external knowledge (Adomako & Nguyen, 2024). This proximity is not merely geographical but also encompasses cognitive and relational dimensions. As a result, organizations can more efficiently identify and connect with potential knowledge, significantly enhancing the speed and efficacy of acquiring innovative resources. From an open innovation perspective, this minimizes external knowledge search costs and broadens the pool of potential partners, enabling organizations to more effectively leverage resources for innovation (Cabrilo et al., 2024). For example, Siemens utilizes its global digital platform to connect R&D centers and partners worldwide, facilitating rapid technological knowledge sharing and collaborative innovation, thus shortening the product development cycle.

Moreover, digital tools foster more frequent and in-depth interactions among organizations. Both explicit and tacit knowledge flows are facilitated through formal collaborative projects and informal exchanges (Shwartz-Asher et al., 2020). These frequent engagements help establish trust and mutual understanding among organizations, promoting knowledge absorption and internalization, and enabling the fusion and innovation of cross-disciplinary insights.

Decentralized information sharing, another function of digitization (Rey et al., 2021), enhances transparency among all participants within the knowledge-sharing network. This fosters trust, reduces knowledge exchange risks, and promotes safer and more effective knowledge sharing (Yang et al., 2021). It diminishes information asymmetry and opportunistic risks in collaboration, creating conditions conducive to secure and efficient knowledge interactions. For example, Haier's "Rendanheyi" model establishes close connections among employees, users, and partners through a digital platform, enabling real-time sharing of demand information and prompt responses. This initiative fosters knowledge collaboration across departments and organizations, leading to more precise and rapid market responses and customized services. Therefore, we propose the following hypothesis:

**Hypothesis 4**. Knowledge digitization has a positive impact on inter-organizational knowledge sharing.

In the digital landscape, collaborative tools and technologies have transformed the design and execution of the innovation process, fostering new forms of cross-organizational collaboration (Urbinati et al., 2022). Digital technology, by facilitating knowledge digitization and sharing, provides enterprises with abundant knowledge resources and collaborative platforms. Adomako and Nguyen (2024) demonstrate that digital tools significantly enhance knowledge sharing efficiency, augmenting technology transfer. This indicates that knowledge digitization, through systematic coding and tool empowerment, lays the foundation for digital knowledge innovation. Additionally, by leveraging digital knowledge resources and collaborative platforms, enterprises can more easily share knowledge with suppliers, partners, research institutions, and others (Cepa, 2021). This inter-organizational knowledge sharing facilitates knowledge flow, integration, and reorganization through networked collaboration, streamlining the digital knowledge innovation chain (Mao et al., 2020). Specifically, knowledge digitization creates the conditions for inter-organizational knowledge sharing, making knowledge acquisition, dissemination, and application more convenient and efficient. In turn, inter-organizational knowledge sharing amplifies knowledge value, promoting digital knowledge innovation through knowledge spillovers, network effects, and synergistic effects. They form a progressive mediating mechanism: knowledge digitization as the foundation, and inter-organizational knowledge sharing as the enhancement. Together, they synergistically transform the potential of digital technology into tangible outcomes for digital knowledge innovation. Hence, we propose that:

**Hypothesis 5**. Knowledge digitization and inter-organizational knowledge sharing operate as serial mediators between digital technology and digital knowledge innovation.

Based on the above analysis, this paper proposes a research model, as shown in Fig. 1.

## 4. Methodology

### *4.1. Sample and data*

In order to ensure rationality, validity, and control common method bias, this study used anonymous completion, designed the questionnaire with a reasonable order and length, and distributed it to digital groups in China engaged in inter-organizational collaboration and collaborative innovation activities. China ranks prominently in global digital competitiveness, particularly in technological innovation, digital transformation, and digital governance. China's digital economy has consistently ranked second globally for several consecutive years (Wan et al., 2024), leading in emerging technologies such as artificial intelligence, cloud computing, big data, and blockchain. Chinese digital groups lead in digital transformation, often possessing extensive practical experience that provides profound insights into cross-organizational collaboration and innovation processes. Moreover, these groups span a wide array of industries, from e-commerce to high-tech sectors, constantly fostering knowledge innovation through diverse digital practices. Consequently, they are ideal participants for studying knowledge digitization and inter-organizational knowledge sharing.

Data were collected through SoJump.com, an online survey platform in mainland China similar to Amazon Mechanical Turk. Participants were randomly selected from the platform's user pool, encompassing various regions across China to ensure sample diversity and representativeness. The questionnaire consists of two parts. The first part is the basic information survey of the respondents, and the second part contains subscales on digital technology, knowledge digitization, inter-organizational knowledge sharing, and digital knowledge innovation. Both items are evaluated using a 7-point Likert scale (1 = very negative and 7 = very positive). Of the 351 questionnaires distributed in this study, 298 valid questionnaires were received (the effective rate is 84.9 %). The characteristics of the respondents and the descriptive statistics of their companies are shown in Fig. 2.

### *4.2. Variable design*

To ensure the reliability, validity, and accuracy of the questionnaire and its results, the measurements of the relevant variables in the study are mainly determined by reviewing through the previous literature and

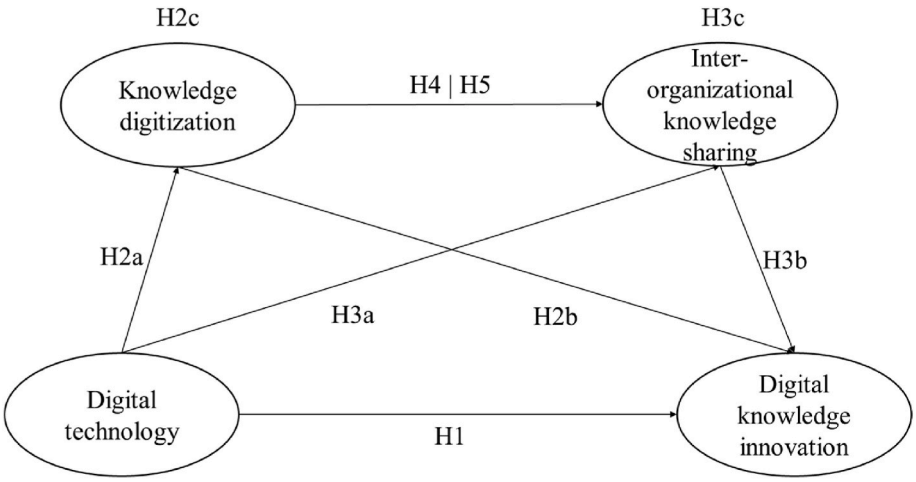



**Fig. 1.** Serial mediation model.

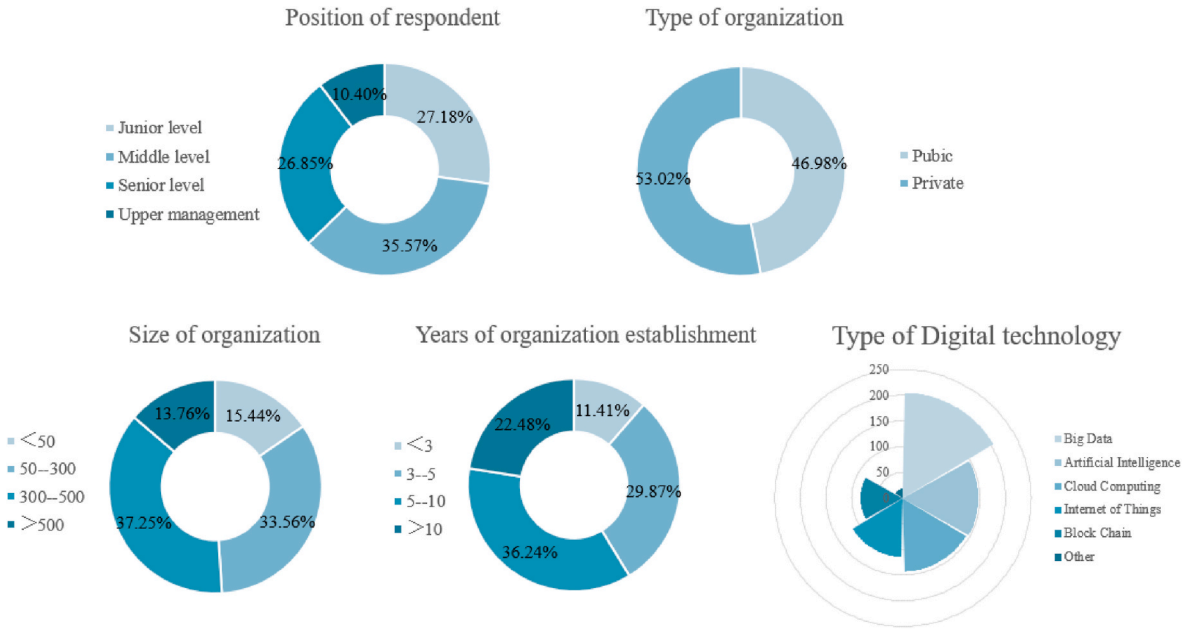



**Fig. 2.** Basic information of respondents (N = 298).

summarizing the views of previous authors, and sorting out the scales of existing studies. Measurement items are attached in the appendix.

Digital technology (independent variable). Referring to the studies such as Ardolino et al. (2018) and Raudeliuniene et al. (2021), this paper mainly determines the measurement items of digital technology from three aspects: organizational management, resource acquisition, and knowledge mining.

Knowledge digitization (mediating variable). Referring to the studies such as Cheng et al. (2023) and Han et al. (2024), this paper determines the measurement items of knowledge digitization from three aspects: codability, convergence and generativity.

Inter-organizational knowledge sharing (mediating variable). Based on the studies such as Forman and van Zeebroeck (2019), Chen et al. (2014) and Ali et al. (2018), this paper determines the measurement items of inter-organizational knowledge sharing from three aspects: knowledge interaction, knowledge reference, and knowledge collaboration.

Digital knowledge innovation (dependent variable). This paper refers to the studies of Hussain et al. (2024) and Wang and Ahmed (2004) to determine the measurement items of digital knowledge innovation from four aspects: knowledge integration, knowledge reconstruction, innovation capability, and innovation performance.

### 4.3. Data analysis

R is a free, open-source statistical language and computing environment that enables users to write and execute the code they need to analyze data (R Core Team, 2023). All methods discussed are primarily implemented in the R language and environment via the lavaan package (Rosseel, 2012), seminr package (Ray et al., 2021). This paper is mainly divided into two steps. Firstly, the reliability and validity of the measurement model are assessed before evaluating the path coefficients or structural relationships of the structural model. Secondly, the structural model is tested using SEM. PLS-SEM is particularly well-suited for theoretical development and models with both reflective and formative constructs (Hair et al., 2021). It can simultaneously estimate causal relationships among constructs and assess their loadings, which is ideal for complex models with latent variables. Therefore, PLS-SEM is considered appropriate to test the hypotheses in this study.

## 5. Results

### 5.1. Common method bias

Following Podsakoff et al. (2003), this paper uses principal component analysis (PCA) to extract four factors with eigenvalues greater than 1. These factors explain 68.810 % of the total variance. The explanatory rate of the first factor is 19.389 %, which does not exceed 40 %. Next, we further test the common method bias using Bifactor confirmatory factor analysis (CFA) by controlling unmeasured latent method construct (ULMC). As shown in Table 1, the fit of the Bifactor model does not improve significantly after adding a method factor to the Four-factor model ($\Delta$CFI = 0.011, $\Delta$TLI = 0.009, $\Delta$RMSEA = 0.005, $\Delta$SRMR = 0.002). Therefore, there is no serious common method bias problem in this paper.

### 5.2. Reliability and validity tests

Table 2 shows that the Cronbach's α of all variables is greater than 0.8, which is above the recommended threshold of 0.7. In addition, none of the 95 % confidence intervals contains 0, indicating that the questionnaire has high reliability and internal consistency. We perform KMO and Bartlett's test on the data to evaluate the suitability of the data for factor analysis. The results show that the KMO test value is 0.953 and all variables pass the Bartlett's test ($p < 0.001$), indicating that the data meet the criteria for factor analysis.

The results of the CFA indicate that all factor loadings exceed 0.7, and all CR values exceed 0.9, which meet the minimum criteria of 0.7. Additionally, the AVE values of all variables are greater than 0.6 and the Rho_A and Rho_C are greater than 0.7 (see Table 2). These results indicate that the measured variables can effectively reflect their latent variables and that the scales have good convergent validity.

Compared with the other models (see Table 1), the four-factor model has the best fit ($\chi^2$/df = 1.871, CFI = 0.960, TLI = 0.954, RMSEA = 0.054, SRMR = 0.033), indicating that the variables have good discriminant validity. In addition, this paper further examines the discriminant validity of the variables through the heterotrait-monotrait (HTMT) ratio (Henseler et al., 2015). As shown in Table 3, the HTMT ratios of the variables are below the strict criterion of 0.85, which further indicates that the scale has good discriminant validity. In summary, the study passes the reliability and validity tests and can be used for further analysis.

### 5.3. Descriptive statistics and correlation analysis

The distribution, mean, standard deviation and correlation coefficient of each variable are shown in Table 4 and Fig. 3. Digital technology shows significant positive correlation with knowledge digitization (r = 0.687, $p < 0.001$), inter-organizational knowledge sharing (r = 0.661, $p < 0.001$), and digital knowledge innovation (r = 0.589, $p < 0.001$). Knowledge digitization is significantly positively correlated with inter-organizational knowledge sharing (r = 0.739, $p < 0.001$), and digital knowledge innovation (r = 0.676, $p < 0.001$). Inter-organizational knowledge sharing and digital knowledge innovation also show significant positive correlation (r = 0.712, $p < 0.001$), indicating a strong correlation between the variables. The results are consistent with the theoretical expectations and preliminarily validate the relevant research hypotheses.

**Table 1**
Results of confirmatory factor analysis.

| Model | Factor | $\chi^2$/df | CFI | TLI | RMSEA | SRMR |
|---|---|---|---|---|---|---|
| Single-factor model | DT + KD + IKS + DKI | 4.878 | 0.815 | 0.796 | 0.114 | 0.068 |
| Two-factor model | DT + KD + IKS, DKI | 3.586 | 0.877 | 0.864 | 0.093 | 0.055 |
| Three-factor model | DT + KD, IKS, DKI | 2.804 | 0.915 | 0.905 | 0.078 | 0.045 |
| Four-factor model | DT, KD, IKS, DKI | 1.871 | 0.960 | 0.954 | 0.054 | 0.033 |
| Bifactor model | DT, KD, IKS, DKI, CMF | 1.711 | 0.971 | 0.963 | 0.049 | 0.031 |

Note: DT, Digital technology; KD, Knowledge digitization; IKS, Inter-organizational knowledge sharing; DKI, Digital knowledge innovation; CMF, Common method factor; +, the factors combined as one.

### 5.4. Hypothesis testing

Before the hypothesis testing, this paper tests the collinearity among the variables via VIF and correlation coefficient. Combined with Tables 4 and 5, it can be seen that the correlation coefficients between the variables are lower than 0.9 and the VIF values are less than 3, indicating that there is no collinearity problem among the variables. In addition, as shown in Fig. 4, the adjusted $R^2$ of each model is between 0.4 and 0.6 ($R^2_{KD} = 0.473$, $R^2_{IKS} = 0.591$, $R^2_{DKC} = 0.563$), $Q^2$ are all greater than 0.3 ($Q^2_{KD} = 0.323$, $Q^2_{IKS} = 0.391$, $Q^2_{DKC} = 0.379$), which indicates that the models are well-fitted, with good explanatory and predictive ability.

#### 5.4.1. Main path test

By including digital technology, knowledge digitization, inter-organizational knowledge sharing, and digital knowledge innovation in the research model, the direct effect relationship between the variables is tested.

As shown in Table 5 and Fig. 4, digital technology has a positive effect on knowledge digitization ($\beta = 0.689$, $t = 15.682$, $p < 0.001$), and knowledge digitization positively impacts digital knowledge innovation ($\beta = 0.276$, $t = 3.993$, $p < 0.001$). Therefore, H2a and H2b are supported. Digital technology has a positive impact on inter-organizational knowledge sharing ($\beta = 0.292$, $t = 3.991$, $p < 0.001$) and inter-organizational knowledge sharing has a positive impact on digital knowledge innovation ($\beta = 0.440$, $t = 6.242$, $p < 0.001$), supporting for H3a and H3b. Knowledge digitization has a positive effect on inter-organizational knowledge sharing ($\beta = 0.540$, $t = 8.040$, $p < 0.001$). Thus, H4 is supported. Next, we further examine the mediation effect.

#### 5.4.2. Mediation effect test

This paper follows the recommendation of Hayes and Preacher (2014), and uses bootstrapping (repeated sampling 5000 times) to test the robustness of the mediating effect between knowledge digitization and inter-organizational knowledge sharing.

As shown in Table 6, the total effect of digital technology on digital knowledge innovation is significant ($\beta = 0.592$, $t = 12.517$, $p < 0.001$), with 95 % CI [0.495, 0.678] excluding 0. Thus, H1 is supported. The total mediating effect is the sum of the three mediation effects ($\beta = 0.482$, $t = 7.306$, $p < 0.001$), and 95 % CI [0.361, 0.622], excluding 0. This indicates that the mediating effect is significant. Digital technology positively affects digital knowledge innovation through knowledge digitization ($\beta = 0.190$, $t = 3.682$, $p < 0.001$), 95 % CI [0.090, 0.295], excluding 0. Thus, H2c is confirmed. The path coefficient of inter-organizational knowledge sharing between digital technology and digital knowledge innovation is 0.128 ($t = 3.375$, $p < 0.001$), 95 % CI [0.061, 0.210], excluding 0. This result is consistent with H3c. The path coefficient of knowledge digitization and inter-organizational knowledge sharing between digital technology and digital knowledge innovation is 0.164 ($t = 0.436$, $p < 0.001$), 95 % CI [0.100, 0.243], excluding 0. This indicates that there is a significant serial mediation effect of knowledge digitization and inter-organizational knowledge sharing. Therefore, H5 is supported.

## 6. Discussion and conclusions

Based on social interaction theory, this paper establishes a serial

**Table 2**
Results of measurement model.

| Variable | Item | Factor loading | Cronbach's α | AVE | CR | Rho_A | Rho_C |
|---|---|---|---|---|---|---|---|
| DT | DT1 | 0.856 | 0.884[0.852, 0.910] | 0.684 | 0.915 | 0.886 | 0.915 |
| | DT2 | 0.854 | | | | | |
| | DT3 | 0.825 | | | | | |
| | DT4 | 0.817 | | | | | |
| | DT5 | 0.780 | | | | | |
| KD | KD1 | 0.826 | 0.893[0.861, 0.919] | 0.701 | 0.921 | 0.894 | 0.921 |
| | KD2 | 0.856 | | | | | |
| | KD3 | 0.848 | | | | | |
| | KD4 | 0.812 | | | | | |
| | KD5 | 0.844 | | | | | |
| IKS | IKS1 | 0.848 | 0.898[0.871, 0.921] | 0.664 | 0.922 | 0.901 | 0.922 |
| | IKS2 | 0.850 | | | | | |
| | IKS3 | 0.801 | | | | | |
| | IKS4 | 0.803 | | | | | |
| | IKS5 | 0.808 | | | | | |
| | IKS6 | 0.775 | | | | | |
| DKI | DKI1 | 0.839 | 0.907[0.885, 0.924] | 0.682 | 0.928 | 0.909 | 0.928 |
| | DKI2 | 0.844 | | | | | |
| | DKI3 | 0.839 | | | | | |
| | DKI4 | 0.818 | | | | | |
| | DKI5 | 0.816 | | | | | |
| | DKI6 | 0.800 | | | | | |

Note: Values in square brackets indicate the 95 % confidence intervals for Cronbach's α.

**Table 3**
Results of HTMT test.

| Variable | DT | KD | IKS | DKI |
|---|---|---|---|---|
| DT | | | | |
| KD | 0.774 | | | |
| IKS | 0.742 | 0.825 | | |
| DKI | 0.659 | 0.751 | 0.790 | |

**Table 4**
Descriptive statistics and correlations with 95 % confidence intervals.

| Variable | Mean | SD | DT | KD | IKS | DKI |
|---|---|---|---|---|---|---|
| DT | 4.89 | 0.920 | 1 | | | |
| KD | 4.951 | 0.936 | 0.687*** [0.622, 0.743] | 1 | | |
| IKS | 5.138 | 1.025 | 0.661*** [0.592, 0.721] | 0.739*** [0.682, 0.786] | 1 | |
| DKI | 5.649 | 0.882 | 0.589*** [0.510, 0.659] | 0.676*** [0.609, 0.733] | 0.712*** [0.652, 0.764] | 1 |

Note: Values in square brackets indicate the 95 % confidence intervals for each correlation; *p < 0.05, **p < 0.01, ***p < 0.001.

mediation model to explain how digital technology influences digital knowledge innovation through knowledge digitization and inter-organizational knowledge sharing. The study shows that digital technology can positively promote digital knowledge innovation, which is consistent with Ilvonen et al. (2018), and validates the applicability of social interaction theory within digital contexts. Furthermore, it is found that knowledge digitization and inter-organizational knowledge sharing significantly mediate the relationship between digital technology and digital knowledge innovation, and these two variables play significant serial mediating roles in the above relationship. This finding enriches social interaction theory and offers a novel theoretical perspective on how digital technologies influence organizational innovation.

### 6.1. Theoretical contributions

This study further unveils the black box of the relationship between digital technology and digital knowledge innovation, elucidating the critical mediating role of inter-organizational knowledge sharing. While previous research acknowledges digital technology's impact on knowledge innovation (Chen & Kim, 2023), it primarily focuses on internal organizational perspectives, overlooking the vital component of inter-organizational interaction. However, knowledge innovation fundamentally entails the synergy between knowledge management and inter-organizational dynamic capabilities (Shafique et al., 2022). We contend that inter-organizational knowledge sharing is a pivotal bridge linking digital technology with knowledge innovation. While digital technology facilitates knowledge flow within organizations, broader synergistic effects and innovative breakthroughs emerge through effective inter-organizational knowledge sharing (Gao et al., 2023). Thus, this study reveals the intrinsic mechanism by which digital technology promotes knowledge innovation, specifically by validating the mediating role of inter-organizational knowledge sharing.

Another contribution is extending knowledge digitization from an internal organizational focus to an inter-organizational level, enriching the application context of knowledge digitization. Existing research primarily centers on internal knowledge digitization's impact on innovation performance. For instance, Cheng et al. (2023) confirm the positive impact of knowledge digitization on firm innovation performance. Han et al. (2024) develop a moderated mediation model encompassing business model innovation and entrepreneurial orientation to explore the relationship between knowledge digitization and high-tech firm performance. However, these studies scarcely address the unique opportunities and challenges of inter-organizational knowledge digitization. This study posits that knowledge digitization should focus not only on internal knowledge conversion and efficiency enhancement but also on leveraging digital means to promote inter-organizational knowledge sharing, achieving broader knowledge synergy and innovation. Knowledge digitization not only aids enterprises in acquiring external knowledge resources, expanding innovation networks, and reducing innovation costs, but also facilitates knowledge flow and collaborative creation across organizational boundaries (Adomako & Nguyen, 2024), fostering more groundbreaking innovative outcomes. This expansion enriches the application context of knowledge digitization while providing a fresh perspective on how inter-organizational knowledge sharing drives innovation.

The third contribution is developing a theoretical framework integrating digital technology, knowledge digitization, inter-organizational knowledge sharing, and digital knowledge innovation, and proposing a

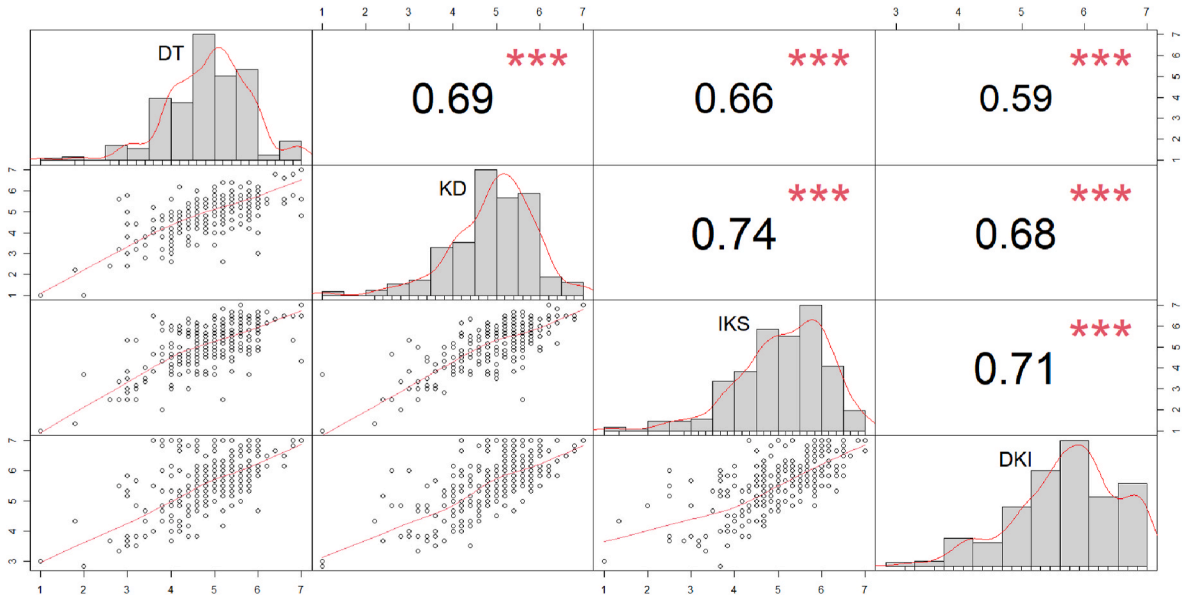



**Fig. 3.** Distribution and correlation of variables.
Note: *p < 0.05, **p < 0.01, ***p < 0.001.

**Table 5**
Results of path test.

| Outcome variable | Predictive variable | β | S.E. | VIF | T-value | 95 % CI | Supported |
|---|---|---|---|---|---|---|---|
| Model 1 | | | | | | | |
| KD | DT (H2a) | 0.689 | 0.044 | 1.000 | 15.682*** | [0.596, 0.769] | Yes |
| Model 2 | | | | | | | |
| IKS | DT (H3a) | 0.292 | 0.073 | 1.904 | 3.991*** | [0.145, 0.432] | Yes |
| | KD (H4) | 0.540 | 0.067 | 1.904 | 8.040*** | [0.407, 0.669] | Yes |
| Model 3 | | | | | | | |
| DKI | DT | 0.110 | 0.081 | 2.115 | 1.352 | [-0.053, 0.263] | |
| | KD (H2b) | 0.276 | 0.069 | 2.621 | 3.993*** | [0.135, 0.409] | Yes |
| | IKS (H3b) | 0.440 | 0.070 | 2.461 | 6.242*** | [0.309, 0.579] | Yes |

Note: 95 % CI, 95 % bias-corrected confidence interval; *p < 0.05, **p < 0.01, ***p < 0.001.

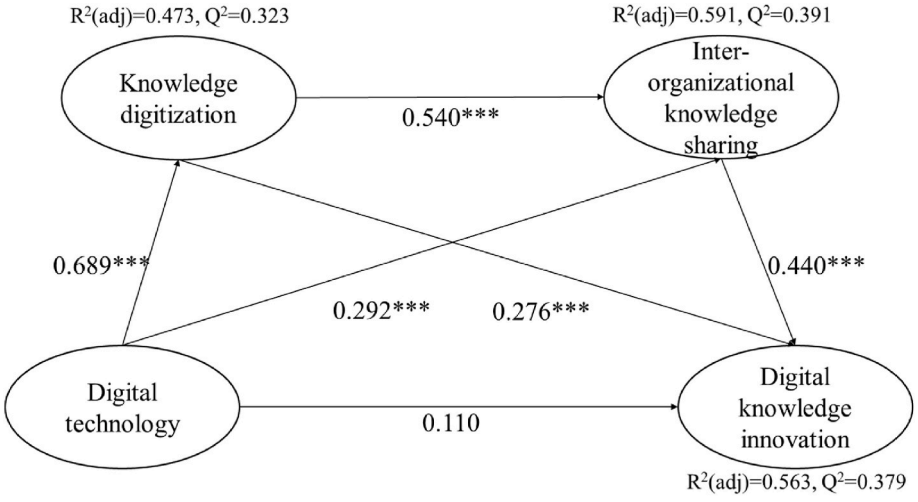



**Fig. 4.** Path testing result.
Note: $R^2$(adj), adjusted $R^2$; *p < 0.05, **p < 0.01, ***p < 0.001.

serial mediation model. Although existing research explores partial relationships among these factors (Chen & Kim, 2023; Du & Wang, 2024; Han et al., 2024), few studies integrate them into a coherent framework for comprehensive analysis. This study identifies the mediating roles of knowledge digitization and inter-organizational knowledge sharing in the relationship between digital technology and digital knowledge innovation and reveals that both are serial mediators. In summary, digital technology reduces information exchange costs, increasing

**Table 6**
Bootstrapping mediation effect test.

| Relationship | Effect | S.E. | T-value | 95 % CI | Supported |
|---|---|---|---|---|---|
| Total effect (H1) | 0.592 | 0.047 | 12.517*** | [0.495, 0.678] | Yes |
| Direct effect | 0.110 | 0.081 | 1.347 | [-0.053, 0.263] | |
| Total indirect effect | 0.482 | 0.066 | 7.306*** | [0.361, 0.622] | |
| DT→KD→DKI (H2c) | 0.190 | 0.052 | 3.682*** | [0.090, 0.295] | Yes |
| DT→IKS→DKI (H3c) | 0.128 | 0.038 | 3.375*** | [0.061, 0.210] | Yes |
| DT→KD→IKS→DKI (H5) | 0.164 | 0.037 | 4.368*** | [0.100, 0.243] | Yes |

Note: 95 % CI, 95 % bias-corrected confidence interval; *p < 0.05, **p < 0.01, ***p < 0.001.

interaction frequency and efficiency, potentially enhancing reciprocity among organizations and fostering knowledge sharing and innovation (Sundaresan, Zhang & Justin, 2020). This provides a more comprehensive explanation of the complex mechanisms by which digital technology impacts on knowledge innovation. This research perspective addresses the gap in existing literature, which primarily focuses on the technology while gives less attention to how technology influences inter-organizational interactions. It offers new theoretical insights into effectively promoting knowledge innovation within a highly dynamic digital environment.

### 6.2. Practical implications

First, organizations need to change the managers' attitude toward new things, which profoundly affects organizational development and competitiveness. Managers should maintain an open and innovative attitude, be willing to try new things, recognize their potential value, adapt to rapid environmental changes, adjust their thinking and behavior, and plan for organizational development and innovation.

Second, organizations should accelerate their digital transformation. Digitalization and emerging technologies serve as guiding forces to consolidate fragmented knowledge, enable information abundance, and drive sustainable organizational growth (Pellizzoni et al., 2020). This includes investing in digital equipment and systems, cultivating digital talent, improving the digital literacy of employees, creating a digital-friendly atmosphere, enhancing digital innovation and continuous improvement capabilities, promoting information communication and knowledge sharing through digital platforms, and leveraging knowledge digitization to enhance the organization's creativity and core competitiveness.

Third, organizations should improve the environment for knowledge interaction. With the advancement of digital technology and economic globalization, inter-organizational communication is becoming increasingly common. Organizations are not closed systems, and strengthening inter-organizational communication is necessary for long-term development. Therefore, organizations should establish effective communication channels, create an open communication atmosphere, encourage partners to openly and frankly exchange opinions and ideas, strengthen knowledge communication and interactive cooperation, promote the mutual flow and continuous sharing of knowledge, maintain updated knowledge, and improve the organization's ability to acquire, integrate, and create knowledge.

### 6.3. Limitations and future research directions

There are some limitations of this study that we will address in future research. First, self-report surveys may introduce common method bias, potentially affecting result reliability. Individual differences among respondents and their interpretations of questions may influence data validity. Employing multiple data sources can mitigate bias associated with a single method. Therefore, future research could adopt mixed methods, combining objective metrics and interviews, to enhance the validity of this study. Second, selecting Chinese digital groups as the sample may lead to potential bias. Although this study employs random sampling, the predominant focus on Chinese digital firms could introduce cultural and market environment biases. Future research could validate the generalizability of these findings by comparing digital groups across other countries and regions.

## CRediT authorship contribution statement

**Qiang Cheng:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis, Conceptualization. **Chun Peng:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Huaxin Wan:** Supervision, Project administration. **Yuanhui Dai:** Supervision, Project administration. **Shuangyang Zhang:** Supervision, Project administration.

## Acknowledgments

This research was supported by the National Social Science Foundation of China (Grant No. 23XGL030). The authors are grateful to the editors and anonymous reviewers for their valuable comments and suggestions, which significantly improved the quality of this article.

## Appendix. Measurement items

| Variables | Items | Contents | Source |
|---|---|---|---|
| Digital technology | DT1 | Organizations broadly apply digital technologies to optimize existing business processes, thereby enhancing operational and production efficiency. | Ardolino et al., 2018; Raudeliuniene et al., 2021 |
| | DT2 | Organizations have successfully harnessed digital technologies to achieve automation, significantly reducing repetitive tasks. | |
| | DT3 | Organizations effectively leverage digital technologies to acquire the latest developments and trends within their industry. | |
| | DT4 | Organizations systematically utilize digital technologies to analyze acquired data, uncovering valuable knowledge and insights. | |
| | DT5 | Organizations maintain a well-curated and regularly updated knowledge repository, ensuring that its content is both accurate and timely. | |
| Knowledge digitization | KD1 | Organizations possess the technological capability to encode and store knowledge effectively. | Cheng et al., 2023; Han et al., 2024 |
| | KD2 | Organizations leverage digital technologies to facilitate the dissemination and reuse of knowledge. | |

(continued on next page)

(*continued*)

| Variables | Items | Contents | Source |
|---|---|---|---|
| | KD3 | Organizations utilize digital technologies to diminish the boundaries of knowledge flow and absorption. | |
| | KD4 | Organizations employ digital means to enhance knowledge collaboration both internally and externally. | |
| | KD5 | Organizations achieve continuous trial-and-error and iterative updates of knowledge through digital means. | |
| Inter-organizational knowledge sharing | IKS1 | We have established formal channels (e.g., collaborative agreements, shared platforms, and regular meetings) for knowledge sharing with other organizations. | Ali et al., 2018; Chen et al., 2014; Forman & van Zeebroeck, 2019 |
| | IKS2 | The company regularly exchanges knowledge, experiences, and perspectives with its business partners. | |
| | IKS3 | We leverage information technology to acquire diverse external information and knowledge regarding competitors and market changes. | |
| | IKS4 | We encourage employees to actively participate in industry events, academic exchanges, and similar activities to acquire the latest knowledge and information. | |
| | IKS5 | The company has strategic goals for mutual learning and knowledge sharing with its business partners. | |
| | IKS6 | The company collaborates with its business partners through joint training initiatives to enhance the knowledge and skills of employees. | |
| Digital knowledge innovation | DKI1 | We are capable of effectively integrating internal and external knowledge resources to support new product development. | Hussain et al., 2024; Wang & Ahmed, 2004 |
| | DKI2 | Compared to our competitors, our recent marketing strategy for products is at the forefront of the market. | |
| | DKI3 | Our innovative products and services are often regarded as highly novel and competitive. | |
| | DKI4 | We continuously refine our business processes to enhance efficiency. | |
| | DKI5 | In terms of bringing new products and services to market, our company typically stands at the forefront of technology. | |
| | DKI6 | We are willing to embrace new ways of working and seek unconventional solutions to enhance project success rates. | |

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.techsoc.2025.102905.

## Data availability

Data will be made available on request.

## References


Adomako, S., & Nguyen, N. P. (2024). Digitalization, inter-organizational collaboration, and technology transfer. *The Journal of Technology Transfer, 49*(4), 1176–1202. https://doi.org/10.1007/s10961-023-10031-z

Agostini, L., Galati, F., & Gastaldi, L. (2019). The digitalization of the innovation process: Challenges and opportunities from a management perspective. *European Journal of Innovation Management, 23*(1), 1–12. https://doi.org/10.1108/EJIM-11-2019-0330

Ahmed, Y. A., Ahmad, M. N., Ahmad, N., & Zakaria, N. H. (2019). Social media for knowledge-sharing: A systematic literature review. *Telematics and Informatics, 37*, 72–112.

Ali, I., Musawir, A. U., & Ali, M. (2018). Impact of knowledge sharing and absorptive capacity on project performance: The moderating role of social processes. *Journal of Knowledge Management, 22*(2), 453–477.

Antonelli, C. (2017). Digital knowledge generation and the appropriability trade-off. *Telecommunications Policy, 41*(10), 991–1002. https://doi.org/10.1016/j.telpol.2016.12.002

Antonelli, C., Crespi, F., & Quatraro, F. (2022). Knowledge complexity and the mechanisms of knowledge generation and exploitation: The European evidence. *Research Policy, 51*(8), Article 104081. https://doi.org/10.1016/j.respol.2020.104081

Appio, F. P., Martini, A., Massa, S., & Testa, S. (2016). Unveiling the intellectual origins of Social Media-based innovation: Insights from a bibliometric approach. *Scientometrics, 108*(1), 355–388. https://doi.org/10.1007/s11192-016-1955-9

Ardolino, M., Rapaccini, M., Saccani, N., Gaiardelli, P., Crespi, G., & Ruggeri, C. (2018). The role of digital technologies for the service transformation of industrial companies. *International Journal of Production Research, 56*(6), 2116–2132. https://doi.org/10.1080/00207543.2017.1324224

Azeem, M., Ahmed, M., Haider, S., & Sajjad, M. (2021). Expanding competitive advantage through organizational culture, knowledge sharing and organizational innovation. *Technology in Society, 66*, Article 101635. https://doi.org/10.1016/j.techsoc.2021.101635

Bednar, K., & Spiekermann, S. (2024). The power of ethics: Uncovering technology risks and positive value potentials in IT innovation planning. *Business & Information Systems Engineering, 66*(2), 181–201. https://doi.org/10.1007/s12599-023-00837-4

Blumer, H. (1986). *Symbolic interactionism: Perspective and method*. University of California Press.

Boeker, W., Howard, M. D., Basu, S., & Sahaym, A. (2021). Interpersonal relationships, digital technologies, and innovation in entrepreneurial ventures. *Journal of Business Research, 125*, 495–507. https://doi.org/10.1016/j.jbusres.2019.09.003

Bouncken, R. B., Kraus, S., & Roig-Tierno, N. (2021). Knowledge- and innovation-based business models for future growth: Digitalized business models and portfolio considerations. *Review of Managerial Science, 15*(1), 1–14. https://doi.org/10.1007/s11846-019-00366-z

Cabrilo, S., Dahms, S., & Tsai, F.-S. (2024). Synergy between multidimensional intellectual capital and digital knowledge management: Uncovering innovation performance complexities. *Journal of Innovation & Knowledge, 9*(4), Article 100568. https://doi.org/10.1016/j.jik.2024.100568

Carbo-Valverde, S., Cuadros-Solas, P., & Rodríguez-Fernández, F. (2020). A machine learning approach to the digitalization of bank customers: Evidence from random and causal forests. *PLoS One, 15*(10), Article e0240362. https://doi.org/10.1371/journal.pone.0240362

Cepa, K. (2021). Understanding interorganizational big data technologies: How technology adoption motivations and technology design shape collaborative dynamics. *Journal of Management Studies, 58*(7), 1761–1799. https://doi.org/10.1111/joms.12740

Chatterjee, S., Chaudhuri, R., Vrontis, D., & Piccolo, R. (2021). Enterprise social network for knowledge sharing in MNCs: Examining the role of knowledge contributors and knowledge seekers for cross-country collaboration. *Journal of International Management, 27*(1), Article 100827. https://doi.org/10.1016/j.intman.2021.100827

Chen, P., & Kim, S. (2023). The impact of digital transformation on innovation performance—the mediating role of innovation factors. *Heliyon, 9*(3), Article e13916. https://doi.org/10.1016/j.heliyon.2023.e13916

Chen, Y.-H., Lin, T.-P., & Yen, D. C. (2014). How to facilitate inter-organizational knowledge sharing: The impact of trust. *Information & Management, 51*(5), 568–578. https://doi.org/10.1016/j.im.2014.03.007

Chen, N., Yang, S., & Li, L. (2023). Research on the influence of digital transformation on enterprise internal control quality. *Journal of Global Information Management, 31*(6), 1–21. https://doi.org/10.4018/JGIM.321187

Chen, H., Zheng, Y., Zeng, S., & Yi, Y. (2024). When is digital technology beneficial toward corporate innovation? A meta-analysis. *IEEE Transactions on Engineering Management, 71*, 5574–5587. https://doi.org/10.1109/TEM.2024.3364705

Cheng, Q., Liu, Y., Peng, C., He, X., Qu, Z., & Dong, Q. (2023). Knowledge digitization: Characteristics, knowledge advantage and innovation performance. *Journal of Business Research, 163*, Article 113915.

Chesbrough, H. (2020). To recover faster from Covid-19, open up: Managerial implications from an open innovation perspective. *Industrial Marketing Management, 88*, 410–413.

Ciarli, T., Kenney, M., Massini, S., & Piscitello, L. (2021). Digital technologies, innovation, and skills: Emerging trajectories and challenges. *Research Policy, 50*(7), Article 104289. https://doi.org/10.1016/j.respol.2021.104289

Dabić, M., Obradović Posinković, T., Vlačić, B., & Gonçalves, R. (2023). A configurational approach to new product development performance: The role of open innovation, digital transformation and absorptive capacity. *Technological Forecasting and Social Change, 194*, Article 122720. https://doi.org/10.1016/j.techfore.2023.122720

Di Vaio, A., Palladino, R., Pezzi, A., & Kalisz, D. E. (2021). The role of digital innovation in knowledge management systems: A systematic literature review. *Journal of Business Research, 123*, 220–231. https://doi.org/10.1016/j.jbusres.2020.09.042

Du, Z., & Wang, Q. (2024). The power of financial support in accelerating digital transformation and corporate innovation in China: Evidence from banking and capital markets. *Financial Innovation, 10*(1), 76. https://doi.org/10.1186/s40854-023-00584-1

Durlauf, S. N., & Ioannides, Y. M. (2010). Social interactions. *Annual Review of Economics, 2*(1), 451–478.

Forman, C., & van Zeebroeck, N. (2019). Digital technology adoption and knowledge flows within firms: Can the Internet overcome geographic and technological distance? *Research Policy, 48*(8), Article 103697. https://doi.org/10.1016/j.respol.2018.10.021

Gao, J.-L., Chen, Y., & Zhang, X.-Q. (2023). Digital technology driving exploratory innovation in the enterprise: A mediated model with moderation. *Systems, 11*(3), 118. https://doi.org/10.3390/systems11030118

Ghasemaghaei, M., Ebrahimi, S., & Hassanein, K. (2018). Data analytics competency for improving firm decision making performance. *The Journal of Strategic Information Systems, 27*(1), 101–113.

Gupta, S., Tuunanen, T., Kar, A. K., & Modgil, S. (2023). Managing digital knowledge for ensuring business efficiency and continuity. *Journal of Knowledge Management, 27*(2), 245–263. https://doi.org/10.1108/JKM-09-2021-0703

Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., Danks, N. P., & Ray, S. (2021). An introduction to structural equation modeling. In *Partial least squares structural equation modeling (PLS-SEM) using R* (pp. 1–29). Springer International Publishing. https://doi.org/10.1007/978-3-030-80519-7_1.

Han, W., Li, X., Zhu, W., Lu, R., & Zu, X. (2024). Knowledge digitization and high-tech firm performance: A moderated mediation model incorporating business model innovation and entrepreneurial orientation. *Technology in Society, 77*, Article 102536. https://doi.org/10.1016/j.techsoc.2024.102536

Hayes, A. F., & Preacher, K. J. (2014). Statistical mediation analysis with a multicategorical independent variable. *British Journal of Mathematical and Statistical Psychology, 67*(3), 451–470.

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science, 43*(1), 115–135. https://doi.org/10.1007/s11747-014-0403-8

Hussain, H., Jun, W., & Radulescu, M. (2024). Innovation performance in the digital divide context: Nexus of digital infrastructure, digital innovation, and E-knowledge. *Journal of the Knowledge Economy*. https://doi.org/10.1007/s13132-024-02058-w

Ilvonen, I., Thalmann, S., Manhart, M., & Sillaber, C. (2018). Reconciling digital transformation and knowledge protection: A research agenda. *Knowledge Management Research and Practice, 16*(2), 235–244. https://doi.org/10.1080/14778238.2018.1445427

Jasimuddin, S. M., Li, J., & Perdikis, N. (2019). An empirical study of the role of knowledge characteristics and tools on knowledge transfer in China-based multinationals. *Journal of Global Information Management, 27*(1), 165–195. https://doi.org/10.4018/JGIM.2019010109

Jiménez-Castillo, D., & Sánchez-Pérez, M. (2013). Nurturing employee market knowledge absorptive capacity through unified internal communication and integrated information technology. *Information & Management, 50*(2–3), 76–86.

Kohli, R., & Melville, N. P. (2019). Digital innovation: A review and synthesis. *Information Systems Journal, 29*(1), 200–223. https://doi.org/10.1111/isj.12193

Lepore, D., Dubbini, S., Micozzi, A., & Spigarelli, F. (2022). Knowledge sharing opportunities for industry 4.0 firms. *Journal of the Knowledge Economy, 13*(1), 501–520. https://doi.org/10.1007/s13132-021-00750-9

Li, P., Guo, X., Wang, F., & Zhang, Q. (2025). Digital transformation and corporate innovation boundaries: Role of supply chain concentration and transparency. *International Review of Financial Analysis, 98*, Article 103922. https://doi.org/10.1016/j.irfa.2025.103922

Liu, B. (2025). How does digital transformation enhance enterprise technological innovation? Evidence from Chinese manufacturing listed companies. *Technology in Society, 82*, Article 102884. https://doi.org/10.1016/j.techsoc.2025.102884

Maione, G., Leoni, G., & Magliacani, M. (2024). Unpacking the knowledge dimensions of digital innovation: Implications for accountability in public and private sectors during extraordinary times. *Journal of Knowledge Management*. https://doi.org/10.1108/JKM-01-2024-0047

Mao, C., Yu, X., Zhou, Q., Harms, R., & Fang, G. (2020). Knowledge growth in university-industry innovation networks – results from a simulation study. *Technological Forecasting and Social Change, 151*, Article 119746. https://doi.org/10.1016/j.techfore.2019.119746

Martínez-Caro, E., Cegarra-Navarro, J. G., & Alfonso-Ruiz, F. J. (2020). Digital technologies and firm performance: The role of digital organisational culture. *Technological Forecasting and Social Change, 154*, Article 119962. https://doi.org/10.1016/j.techfore.2020.119962

Mead, G. H., Huebner, C. W. M. A. E. by D. R., & Joas, H. (Eds.). (2015). *Mind, self, and society: The definitive edition*. University of Chicago Press.

Nahapiet, J., & Ghoshal, S. (1998). Social capital, intellectual capital, and the organizational advantage. *Academy of Management Review, 23*(2), 242. https://doi.org/10.2307/259373

Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital innovation management: Reinventing innovation management research in a digital world. *MIS Quarterly, 41*(1), 223–238.

Nonaka, I., & Takeuchi, H. (1995). *The knowledge-creating company: How Japanese companies create the dynamics of innovation*. York, NY: Oxford University PressNew. https://doi.org/10.1093/oso/9780195092691.001.0001

Orlando, B., Mazzucchelli, A., Usai, A., Nicotra, M., & Paoletti, F. (2020). Are digital technologies killing future innovation? The curvilinear relationship between digital technologies and firm's intellectual property. *Journal of Intellectual Capital, 22*(3), 587–609. https://doi.org/10.1108/JIC-03-2020-0078

Otioma, C. (2023). IT capability, organisational learning and innovation performance of firms in Kenya. *Journal of the Knowledge Economy, 14*(3), 3489–3517. https://doi.org/10.1007/s13132-021-00886-8

Ozdemir, S., & Gupta, S. (2021). Inter-organizational collaborations for social innovation and social value creation: Towards the development of new research agenda and theoretical perspectives. *Industrial Marketing Management, 97*, 134–144. https://doi.org/10.1016/j.indmarman.2021.06.013

Pellizzoni, E., Trabucchi, D., Frattini, F., Buganza, T., & Di Benedetto, A. (2020). Leveraging stakeholders' knowledge in new service development: A dynamic approach. *Journal of Knowledge Management, 24*(2), 415–438. https://doi.org/10.1108/JKM-10-2019-0532

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879

R Core Team. (2023). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org.

Rahul, K., Banyal, R. K., & Arora, N. (2023). A systematic review on big data applications and scope for industrial processing and healthcare sectors. *Journal of Big Data, 10*(1), 133. https://doi.org/10.1186/s40537-023-00808-2

Raudeliuniene, J., Albats, E., & Kordab, M. (2021). Impact of information technologies and social networks on knowledge management processes in Middle Eastern audit and consulting companies. *Journal of Knowledge Management, 25*(4), 871–898. https://doi.org/10.1108/JKM-03-2020-0168

Ravichandran, T., & Giura, S. I. (2019). Knowledge transfers in alliances: Exploring the facilitating role of information technology. *Information Systems Research, 30*(3), 726–744.

Ray, S., Danks, N., & Calero Valdez, A. (2021). SEMinR: Domain-specific language for building, estimating, and visualizing structural equation models in R. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3900621

Rehman, F. U., Ismail, H., Al Ghazali, B. M., Asad, M. M., Shahbaz, M. S., & Zeb, A. (2021). Knowledge management process, knowledge based innovation: Does academic researcher's productivity mediate during the pandemic of covid-19? *PLoS One, 16*(12), Article e0261573. https://doi.org/10.1371/journal.pone.0261573

Rey, A., Panetti, E., Maglio, R., & Ferretti, M. (2021). Determinants in adopting the Internet of Things in the transport and logistics industry. *Journal of Business Research, 131*, 584–590.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software, 48*, 1–36. https://doi.org/10.18637/jss.v048.i02

Ryan, S., & O'Connor, R. V. (2013). Acquiring and sharing tacit knowledge in software development teams: An empirical study. *Information and Software Technology, 55*(9), 1614–1624. https://doi.org/10.1016/j.infsof.2013.02.013

Shafique, I., Kalyar, M. N., Shafique, M., Kianto, A., & Beh, L.-S. (2022). Demystifying the link between knowledge management capability and innovation ambidexterity: Organizational structure as a moderator. *Business Process Management Journal, 28*(5/6), 1343–1363. https://doi.org/10.1108/BPMJ-11-2021-0713

Shwartz-Asher, D., Chun, S., Adam, N. R., & Snider, K. L. (2020). Knowledge sharing behaviors in social media. *Technology in Society, 63*, Article 101426.

Sundaresan, S., Zhang, Z., & Justin. (2020). Knowledge-sharing rewards in enterprise social networks: Effects of learner types and impact of digitisation. *Enterprise Information Systems, 14*(5), 661–679. https://doi.org/10.1080/17517575.2020.1737884

Švarc, J., Dabić, M., & Tekavčič, M. (2024). Controversies surrounding digitalization: Discussions in four key domains. In M. Dabić, J. Švarc, & T. U. Daim (Eds.), *Digital entrepreneurship in science, technology and innovation* (pp. 353–372). Edward Elgar Publishing. https://doi.org/10.4337/9781035311422.00022.

Takko, T., Bhattacharya, K., Lehto, M., Jalasvirta, P., Cederberg, A., & Kaski, K. (2023). Knowledge mining of unstructured information: Application to cyber domain. *Scientific Reports, 13*(1). https://doi.org/10.1038/s41598-023-28796-6. Article 1.

Ting, D. S. W., Carin, L., Dzau, V., & Wong, T. Y. (2020). Digital technology and COVID-19. *Nature Medicine, 26*(4), 459–461. https://doi.org/10.1038/s41591-020-0824-5

Tønnessen, Ø., Dhir, A., & Flåten, B.-T. (2021). Digital knowledge sharing and creative performance: Work from home during the COVID-19 pandemic. *Technological Forecasting and Social Change, 170*, Article 120866. https://doi.org/10.1016/j.techfore.2021.120866

Urbinati, A., Manelli, L., Frattini, F., & Bogers, M. L. A. M. (2022). The digital transformation of the innovation process: Orchestration mechanisms and future research directions. *Innovation, 24*(1), 65–85. https://doi.org/10.1080/14479338.2021.1963736

Volz, F., Münch, C., Küffner, C., & Hartmann, E. (2025). Digital ecosystems and their impact on organizations—a dynamic capabilities approach. *International Journal of Management Reviews*, 1–12. https://doi.org/10.1111/ijmr.12396

Wan, X., Wang, Y., & Zhang, W. (2024). The spatial and temporal situation of China's digital technology innovation and its influencing factors. *PLoS One, 19*(1), Article e0297401. https://doi.org/10.1371/journal.pone.0297401

Wang, C. L., & Ahmed, P. K. (2004). The development and validation of the organisational innovativeness construct using confirmatory factor analysis. *European Journal of Innovation Management, 7*(4), 303–313. https://doi.org/10.1108/14601060410565056

Xia, Y., & Johar, M. G. M. (2025). How external factors influence organisational digital innovation: Evidence from China. *Technology in Society, 81*, Article 102802. https://doi.org/10.1016/j.techsoc.2024.102802

Xie, X., Han, Y., Anderson, A., & Ribeiro-Navarrete, S. (2022). Digital platforms and SMEs' business model innovation: Exploring the mediating mechanisms of capability reconfiguration. *International Journal of Information Management, 65*, Article 102513. https://doi.org/10.1016/j.ijinfomgt.2022.102513

Yang, X., Ye, H. J., & Wang, X. (2021). Social media use and work efficiency: Insights from the theory of communication visibility. *Information & Management, 58*(4), Article 103462.

Zhou, L., Mao, H., Zhao, T., Wang, V. L., Wang, X., & Zuo, P. (2022). How B2B platform improves Buyers' performance: Insights into platform's substitution effect. *Journal of Business Research, 143*, 72–80. https://doi.org/10.1016/j.jbusres.2022.01.060

Zhou, Q., Wang, S., Wang, L., & Xu, W. (2025). Knowledge governance and innovation ambidexterity in the platform context: Exploring the role of knowledge transformation. *Journal of Knowledge Management, 29*(4), 1301–1329. https://doi.org/10.1108/JKM-03-2024-0256

Zou, H., Ji, H., Qin, H., Guo, Y., & Yang, X. (2022). Research on the strategy evolution of knowledge innovation in an enterprise digital innovation ecosystem: Kinetic and potential perspectives. *IEEE Access, 10*, 78764–78779. https://doi.org/10.1109/ACCESS.2022.3194071